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SUB-SOIL INVESTIGATION REPORT FOR PROPOSED ON 06 STORIED RESIDENTIAL BUILDING AT HOWLADARPARA, SYLHET DAG # SA-56, BS-1744, KHA # SA-109, BS-1756 J.L # SA-77, BS-72, MOUJA- BRAMMONSHASHON

Verification Code: JgkQS88P
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CLIENT

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CONDUCTED BY
CENTER FOR RESEARCH TESTING & CONSULTANCY
DEPARTMENT OF CIVIL & ENVIRONMENTAL ENGINEERING
SHAHJALAL UNIVERSITY OF SCIENCE & TECHNOLOGY, SYLHET

Sample Collected by
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Introduction

A reasonably accurate conception about the sub-soil parameters of any project site is an essential prerequisite for proper planning and designing the foundation of the structure so that the same after its construction would remain safe and serviceable. A detailed study on the sub-surface elements in this regard would necessary to select proper type of foundation.

Objectives

The purpose of sub-soil investigation program is to obtain the following information

1 To classify the formation of soil layers

2 To estimate the safe bearing capacities of the foundation at different soil layers

3 To suggest the safe and appropriate type of foundations.

Scope of work

Centre for research, testing and consultancy (CRTC), Department of Civil and Environmental Engineering, Shahjalal University of Science & Technology (SUST), Sylhet; execute only the laboratory tests and other computational works with the supplied disturbed and/or undisturbed samples to determine the geotechnical properties of the soil.

Table – 1: Items of the supplied samples

No	Name	Bore hole no	Depth	Quantities
01	Disturbed sample	01	0-50 ft	10
02	Disturbed sample	02	0-55 ft	11
03	Disturbed sample	03	0-50 ft	10
04	Disturbed sample	04	0-45ft	09
05	Disturbed sample	05	0-50ft	10

Laboratory Classifications

The disturbed soil samples, supplied by the client were thoroughly examined and classified roughly (color and type) and noted in accordance with bore hole (BH) and depth (Please see enclosed Tables).

Laboratory Works & Analysis

In addition to visual examination of the sample, Bearing Capacity of soil samples including skin friction & end bearing were calculated.

Bearing capacity for shallow foundation

The N value gives useful information with regards to consistency of cohesive soil and relative density of cohesion less soil. These relationships are given in the following Tables for cohesive and cohesion less soil, respectively.

Table 2: Relation of N-value with the Consistency and the Unconfined Compressive Strength of Cohesive Soils (after Bowles 1997)

N-values	Consistency	Unconfined Compressive Strength (q_u)	
		tsf	kPa
0-2	Very soft	0 to 0.25	0 to 25
3-5	Soft	0.25 to 0.50	25 to 50
6-9	Medium	0.50 to 1	50 to 100
10-16	Stiff	1 to 2	100 to 200
17-30	Very stiff	2 to 4	200 to 400
>30	Hard	>4	>400

Table 3: N and ϕ Relation to Relative Density for Fine Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
1 – 2	Very Loose	0 – 15	26 – 28	70 – 100
3 – 6	Loose	15 – 35	28 – 30	90 – 115
7 – 15	Medium	35 – 65	30 – 34	110 – 130
16 – 30	Dense	65 – 85	33 – 38	110 – 140
>30	Very Dense	>85	>50	130 – 150

Table 4: N and ϕ Relation to Relative Density for Medium Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
2 – 3	Very Loose	0 – 15	27 – 28	70 – 100
4 – 7	Loose	15 – 35	30 – 32	90 – 115
8 – 20	Medium	35 – 65	32 – 36	110 – 130
21 – 40	Dense	65 – 85	36 – 42	110 – 140

Table 5: N and ϕ Relation to Relative Density for Coarse Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
3 – 6	Very Loose	0 – 15	28 – 30	70 – 100
5 – 9	Loose	15 – 35	30 – 34	90 – 115
10 – 25	Medium	35 – 65	33 – 40	110 – 130
26 – 40	Dense	65 – 85	40 – 50	110 – 140
>40	Very Dense	>85	>50	130 – 150

Bearing capacity for deep foundation

Foundations, preferably piles are considered to transmit heavy load from superstructure to deep and hard strata. As the formation of soil strata consists of cohesive and non-cohesive type, the evaluation criteria for the determination of skin friction and the end bearing capacities of piles also vary. Details of the evaluation are provided in the following sub-headings.

Table 6: Formula Used to Determine Bearing Capacity of Soil

Shallow Foundation	
For Cohesive Soil	
$q_a = q_u + \gamma \frac{D}{3}$ $q_a = q_u$(Common Practice)	Where, q_a=Allowable bearing capacity, q_u= Unconfined Compressive Strength, γ= Unit Weight of soil, D= Depth of footing
For Cohesion less Soil	
$q_{ult} = 30N$; $q_a = q_{ult}/FS$	q_{ult}= Ultimate bearing capacity in kPa, N = SPT Number, FS= Factor of Safety
Source: Bowles, J. E., "Physical and Geotechnical Properties of Soils", 2nd Edition	
Deep Foundation	
For Cohesive Soil	
$f_{su} = \alpha C_u$; $q_{pu} = N_c C_u$	α = Adhesion factor, f_{su} = Ultimate skin friction, C_u = Undrained Shear Strength, q_{pu} = Ultimate end bearing of pile, N_c = Bearing capacity factor for deep foundation=9
For Cohesion less Soil	
$f_{su} = N/50\ tsf$; $q_{pu} = 4N\ tsf$	f_{su} = Ultimate skin friction, q_{pu} = Ultimate end bearing of pile, N = SPT Number
Source: Rajapakse RA (2008), "Geotechnical Engineering Calculations and Rules of Thumb"	

Based on the above formula, the important parameters for foundation design have been determined and are given in **Table 7**, **Table 8, Table 9, Table 10 and Table 11** respectively.

Table 7: The allowable bearing capacity for shallow foundation in Tsf (FS=2.5)

Bore Hole No.	Depth	Design N	Clay						Sand
			Continuous			Square(Isolated)			
	(ft)	-	D/B=0.25	D/B=0.5	D/B=1	D/B=0.25	D/B=0.5	D/B=1	
BH01	2.5-5.0	2	0.278	0.293	0.319	0.329	0.355	0.386	0.300
	5.0-10	3	0.388	0.408	0.445	0.458	0.495	0.539	0.418
BH02	2.5-5.0	10	1.206	1.269	1.383	1.425	1.539	1.674	1.300
	5.0-10	10	1.104	1.161	1.266	1.304	1.409	1.532	1.190
BH03	2.5-5.0	10	1.160	1.220	1.330	1.370	1.480	1.610	1.250
	5.0-10	13	1.520	1.599	1.743	1.795	1.939	2.110	1.638
BH04	2.5-5.0	4	0.510	0.537	0.585	0.603	0.651	0.708	0.550
	5.0-10	4	0.460	0.484	0.527	0.543	0.587	0.638	0.496
BH05	2.5-5.0	5	0.580	0.610	0.665	0.685	0.740	0.805	0.625
	5.0-10	5	0.536	0.564	0.615	0.633	0.684	0.744	0.578

Table 8: The allowable skin friction for Precast driven piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.049	0.096	0.138	0.096	0.118					
10	0.074	0.230	0.218	0.118	0.138					
15	0.049	0.218	0.261	0.096	0.096					
20	0.118	0.118	0.240	0.074	0.096					
25	0.118	0.196	0.240	0.192	0.168					
30	0.226	0.240	0.230	0.200	0.174					
35	0.261	0.268	0.240	0.200	0.246					
40	0.241	0.401	0.230	0.216	0.321					
45	0.208	0.334	0.308	0.208	0.208					
50	0.200	0.200	0.200		0.200					
55		0.192	0.192							

Table 9: The allowable end bearing for Precast driven piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.450	0.900	1.350	0.900	1.125					
10	0.675	3.150	2.250	1.125	1.350					
15	0.450	2.250	2.700	0.900	0.900					
20	1.125	1.125	3.825	0.675	0.900					
25	1.125	2.025	3.825	38.400	33.600					
30	3.600	3.825	3.150	40.000	1.800					
35	2.700	4.500	3.825	40.000	3.375					
40	4.050	6.750	3.150	43.200	5.400					
45	41.600	5.625	5.175	41.600	41.600					
50	40.000	40.000	40.000		40.000					
55		38.400	38.400							

Table 10: The allowable skin friction for Bored piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.032	0.062	0.090	0.062	0.077					
10	0.048	0.149	0.141	0.077	0.090					
15	0.032	0.141	0.170	0.062	0.062					
20	0.077	0.077	0.156	0.048	0.062					
25	0.077	0.127	0.156	0.125	0.109					
30	0.147	0.156	0.149	0.130	0.113					
35	0.170	0.174	0.156	0.130	0.160					
40	0.156	0.261	0.149	0.140	0.209					
45	0.135	0.217	0.200	0.135	0.135					
50	0.130	0.130	0.130		0.130					
55		0.125	0.125							

Table 11: The allowable end bearing for Bored piles in Tsf (FS=3.0)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.203	0.405	0.608	0.405	0.506					
10	0.304	1.418	1.013	0.506	0.608					
15	0.203	1.013	1.215	0.405	0.405					
20	0.506	0.506	1.721	0.304	0.405					
25	0.506	0.911	1.721	17.280	15.120					
30	1.620	1.721	1.418	18.000	0.810					
35	1.215	2.025	1.721	18.000	1.519					
40	1.823	3.038	1.418	19.440	2.430					
45	18.720	2.531	2.329	18.720	18.720					
50	18.000	18.000	18.000		18.000					
55		17.280	17.280							

CONCLUSION AND RECOMMENDATION

On the basis of analysis and information provided by the owner following recommendations are suggested for the proposed six storied building.

Soil profile scenario - Overall soil information of the investigated site is quite regular in between the boreholes and vertical consistency in the soil formation at different layers was observed. Nearly grey soft clay soil with organic black soil scenario has been observed among the boreholes. For borehole 1 to 5 at top surface, there is layer of soft clay with low bearing capacity clay in most of the site. There is a dense and hard clay layer below 40ft from ground level in most of the site. There is hard sand layer below 40ft in the site.

Foundation options - In this situation, the foundation should be selected considering both the dimension and design load for the proposed building. Since hard sandy soil layer is found at 40 ft, pilling can be adopted up to required depth (up to 45ft). The bearing capacity for shallow foundation which is given in **table 7** is based on existing soil conditions. However, client needs to reassess the improved bearing capacity of the prepared ground after implementing soil improvement technique in the top layer as mentioned above.

Safety provisions - Since the proposed site is composed of alternate layers of soft cohesive and granular soil, it is reasonably vulnerable to differential settlement. Therefore, it is strongly recommended to check the **settlement** of the site considering building load while designing either shallow or deep foundation. Obviously **pile load test** covering the entire project area must be carried out for confirming the design parameters of the service pile (*if pile foundation is considered*). Moreover, the earthquake vulnerability of the project area as well as the quality management of the construction work including workmanship should be taken into account for the safe design of shallow or deep foundation.

Attachments - The layout of the bore holes of the investigated site and the bore-log profiles with SPT values is shown in the Appendix. Client performed no special laboratory test for the proposed site.

Special notes - These suggestions are absolutely tentative and selection of type, depth and dimensions of foundation would be decided by the Consultant or Design Engineer. It is also to be noted that the above recommendation is restricted only to the existing soil condition as client provided.

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